Ali Pakniyat

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⋄ https://scholar.google.ca/citations?user=HbtfyfwAAAAJ

SUMMARY

Senior Controls & Autonomy Researcher with over 15 years of experience designing optimal, robust, and stochastic control algorithms for autonomous robotics, and multi-agent systems. Proven track record of leading complex R&D projects, securing cross-institutional funding, and transitioning theoretical mathematical frameworks into applied technologies.

TECHNICAL SKILLS

Domain Expertise: Nonlinear & Hybrid Systems, Optimal & Robust Control, Stochastic Systems, Multi-Agent/Swarm Robotics, Collision Avoidance, Path Planning, Guidance, Navigation, and Control (GNC)
Programming & Tools: MATLAB/Simulink, C++, Python
Application Areas: Vehicle Electrification, Autonomous Driving, Unmanned Aerial Vehicles (UAVs), Micro Electrical-Mechanical Systems (MEMS), Large-Scale Networks

PROFESSIONAL EXPERIENCE

University of Alabama, Tuscaloosa, AL — *2021 - present*
Assistant Professor & Director of $\mathcal{M}^3\mathcal{AC}$ Lab, Department of Mechanical Engineering

- Direct the Multi-Modal Multi-Agent Control ($\mathcal{M}^3\mathcal{AC}$) Lab, managing a team of 6+ researchers developing optimality- and safety-critical control algorithms for autonomous vehicles and multi-robot systems.
- Serve as Principal Investigator on a $650k NSF collaborative project developing advanced state and covariance steering methods for nonlinear and hybrid stochastic systems.
- Formulated tube-based tightened exponential barrier function controllers to guarantee robust collision avoidance in uncertain & nonlinear multi-agent networks.
- Developed graphon mean-field control strategies to optimize probability distributions and collective behavior in large-scale multi-agent and swarm networks.
- Bridged stochastic control and Generative AI by synthesizing time-reversal control frameworks to steer complex stochastic systems, inspired by diffusion models.
- Pioneered control-theoretic modeling and optimization techniques for Transformers and Large Language Models (LLMs) to analyze and enhance foundation model dynamics and control.
- Developed autonomous guidance and control algorithms for robotic thrombectomy, combining stochastic control with real-time trajectory optimization to improve precision and safety in minimally invasive neurovascular procedures.

Georgia Institute of Technology, Atlanta, GA — *2019 - 2021*
Postdoctoral Research Fellow, Institute for Robotics and Intelligent Machines (IRIM)

- Developed sampling-based algorithms to solve Feynman-Kac forward-backward SDEs, significantly improving scalable computation of value functions for high-dimensional stochastic optimal control.
- Co-developed control and motion planning algorithms (e.g., forward-backward RRTs) for stochastic autonomous systems, advancing guidance and navigation architectures.

University of Michigan, Ann Arbor, MI — *2017 - 2019*
Postdoctoral Research Fellow, Department of Mechanical Engineering

- Engineered a novel convex duality approach to the optimal control of killed Markov processes, utilizing Sums of Squares (SoS) polynomials to translate theoretical stochastic controls into computationally tractable control algorithms.

McGill University, Montreal, Canada — *2011 - 2017*
Doctoral Researcher & Lecturer, Department of Electrical and Computer Engineering

- Extended the Minimum Principle and Dynamic Programming for hybrid systems via variational approaches.
- Designed, modeled, and experimentally validated hybrid control strategies for seamless dual-brake transmissions in electric vehicles, directly leading to a granted US Patent (US 9,702,438 B2).

EDUCATION

McGill University, Montreal, Canada — *2011 – 2016*
Doctor of Philosophy, Electrical Engineering — **GPA: 4.0** / 4

Sharif University of Technology, Tehran, Iran — *2008 – 2010*
Master of Science, Mechanical Engineering — **GPA: 4.0** / 4

Shiraz University, Shiraz, Iran — *2004 – 2008*
Bachelor of Science, Mechanical Engineering — **GPA: 3.94** / 4

SELECTED HONOURS AND AWARDS

- ▷ National Science Foundation (NSF) Award (PI): \$650k Collaborative Research — 2026–2029
- ▷ Automotive Partnership Canada (APC) – NSERC Grant Contributor — 2012–2016
- ▷ Canadian Marconi Graduate Award — 2017
- ▷ Graduate Excellence Award in Engineering – McGill University — 2011–2014

SELECTED PUBLICATIONS & PATENTS

Full list of 40+ publications available on Google Scholar. Citations: 650+, h-index: 16

Patent

[P1] B. Boulet, M. S. R. Mousavi, H. V. Alizadeh, and **A. Pakniyat**, "Seamless Transmission Systems and Methods for Electric Vehicles," Jul. 11 2017, US Patent US 9,702,438 B2

Selected Journal Papers

[J6] A. Koulong and **A. Pakniyat**, "Robust Multi-Agent Safety via Tube-Based Tightened Exponential Barrier Functions," *Nonlinear Analysis: Hybrid Systems*, vol. 61, p. 101717, 2026

[J5] **A. Pakniyat**, "A Graphon Mean Field Convex Duality Approach to Shaping the Terminal Probability Distribution of a Network of Nonlinear Multi-Agent Systems," *Journal of Systems Science and Complexity*, vol. 38, no. 1, pp. 349–368, 2025

[J4] K. P. Hawkins, **A. Pakniyat**, E. Theodorou, and P. Tsiotras, "Solving Feynman-Kac Forward Backward SDEs Using McKean-Markov Branched Sampling," *IEEE Transactions on Automatic Control*, vol. 69, no. 9, pp. 5695–5710, 2024

[J3] T. Yasini and **A. Pakniyat**, "Hybrid Optimal Control of a Flying+Sailing Drone," *ASME Letters in Dynamic Systems and Control*, vol. 3, no. 3, pp. 031 008–1 – 031 008–7, 2023

[J2] K. P. Hawkins, **A. Pakniyat**, and P. Tsiotras, "Value Function Estimators for Feynman–Kac Forward–Backward SDEs in Stochastic Optimal Control," *Automatica*, vol. 158, p. 111281, 2023

[J1] M. S. R. Mousavi, **A. Pakniyat**, T. Wang, and B. Boulet, "Seamless Dual Brake Transmission For Electric Vehicles: Design, Control and Experiment," *Mechanism and Machine Theory*, vol. 94, pp. 96–118, 2015

Selected Conference Papers

[C5] Y. Mei, A. Taghvaei, and **A. Pakniyat**, "A Time-Reversal Control Synthesis for Steering the State of Stochastic Systems," in *Proceedings of the IEEE Conference on Decision and Control, Rio de Janeiro, Brazil*, 2025

[C4] A. Koulong and **A. Pakniyat**, "Distributed Adaptive Consensus with Obstacle and Collision Avoidance for Networks of Heterogeneous Multi-Agent Systems," in *Proceedings of the IEEE American Control Conference, Denver, CO, USA*. IEEE, 2025, pp. 3602–3607

[C3] K. P. Hawkins, **A. Pakniyat**, E. Theodorou, and P. Tsiotras, "Forward-Backward Rapidly-Exploring Random Trees for Stochastic Optimal Control," in *Proceedings of the IEEE Conference on Decision and Control, Austin, USA*, 2021, pp. 912–917

[C2] **A. Pakniyat** and P. Tsiotras, "Steering the State of Linear Stochastic Systems: A Constrained Minimum Principle Formulation," in *Proceedings of the IEEE American Control Conference, New Orleans, USA*, 2021, pp. 1300–1305

[C1] **A. Pakniyat** and P. E. Caines, "On the Stochastic Minimum Principle for Hybrid Systems," in *Proceedings of the 55th IEEE Conference on Decision and Control, Las Vegas, NV, USA*, 2016, pp. 1139–1144

LEADERSHIP, MENTORING & SERVICE

- **Team Leadership:** Mentored and supervised 10+ Ph.D. and M.Sc. engineers. Lab alumni have successfully transitioned to R&D roles at NASA Marshall Space Flight Center, General Motors, Toyota Research Institute (TRI), and the Defense Intelligence Agency.
- **Conference Organization & Committees:** Served on the Program Committees for the ACM International Conference on Hybrid Systems: Computation and Control (HSCC) and the International Conference on Adaptive and Self-Adaptive Systems (ADAPTIVE).
- **Subject Matter Expert / Reviewer:** Regular technical reviewer for National Science Foundation (NSF) funding panels and highly-ranked journals, including *IEEE TAC*, *Automatica*, and *Nonlinear Analysis: Hybrid Systems.*
- **Technical Speaker:** Delivered over 20 invited talks on autonomous systems, optimal control, and mean-field games at top universities and industry seminars (e.g., Stanford, UC Irvine).
- **Academic & Professional Leadership:** Chaired the IEEE Control Systems Society (Southeast Michigan Section) and served on the Mechanical Engineering Faculty Search Committee at the University of Alabama.